

Lecture #13: The Bivariate and Multivariate Normal Distributions

MATH 321 – Mathematical Statistics

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Normal Distribution

We've now dealt with the single variable Normal distribution a lot.

$$X \sim \text{Normal}(\mu, \sigma^2)$$
$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}$$

How might we generalize this to several variables?

The Bivariate Normal Distribution

We could use independence.

Suppose that:

$$X \sim \text{Normal}(\mu_1, \sigma_1^2)$$

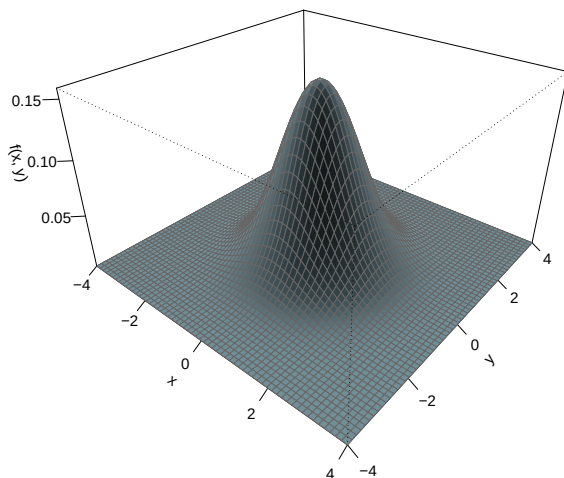
$$Y \sim \text{Normal}(\mu_2, \sigma_2^2)$$

If X and Y are independent, then:

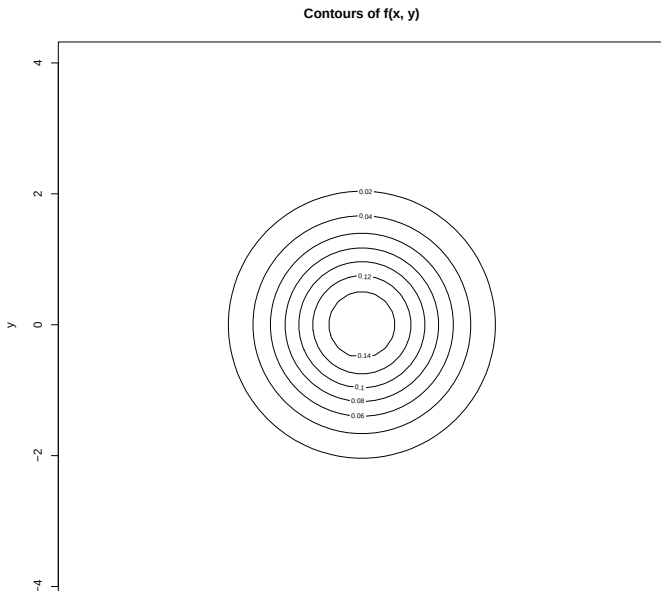
$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_1\sigma_2} \exp \left\{ -\frac{1}{2} \left[\left(\frac{x - \mu_1}{\sigma_1} \right)^2 + \left(\frac{y - \mu_2}{\sigma_2} \right)^2 \right] \right\}$$

The Bivariate Normal Distribution

Independent Standard Normals

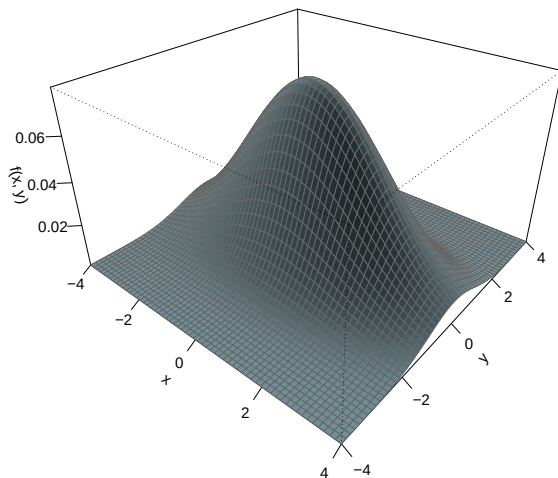


The Bivariate Normal Distribution

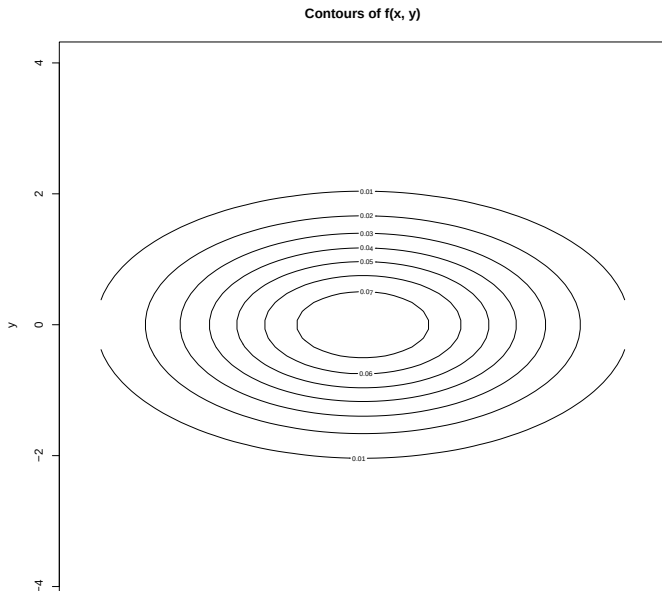


The Bivariate Normal Distribution

Independent Normals with $\mu_x = 2$

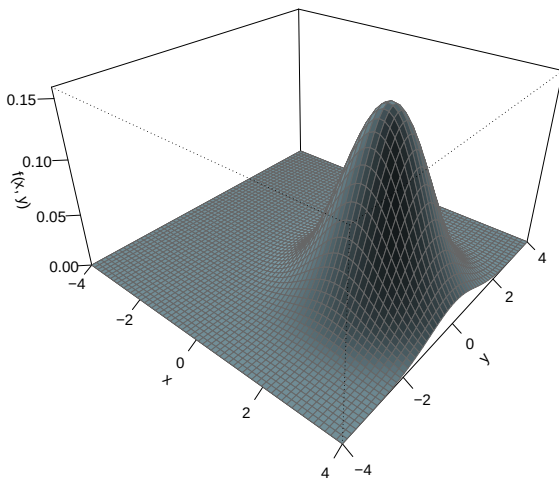


The Bivariate Normal Distribution

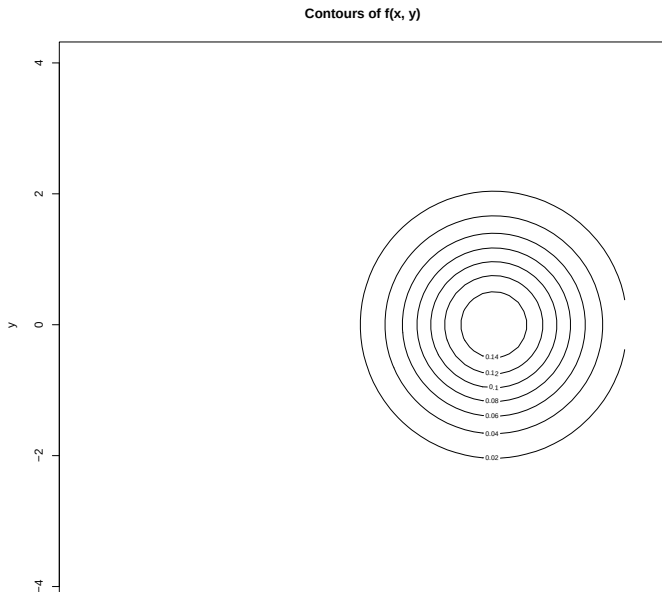


The Bivariate Normal Distribution

Independent Normals with shifted mean

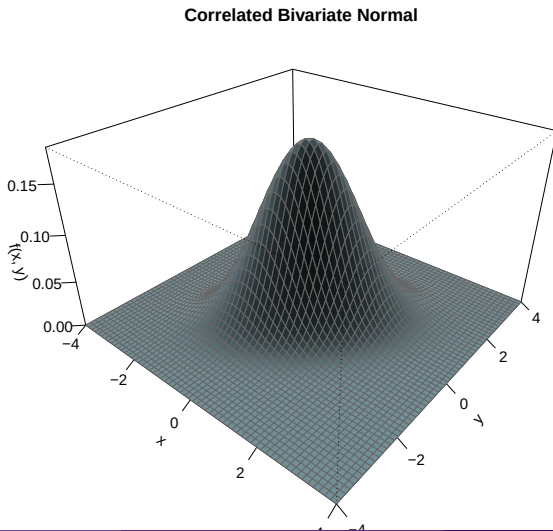


The Bivariate Normal Distribution



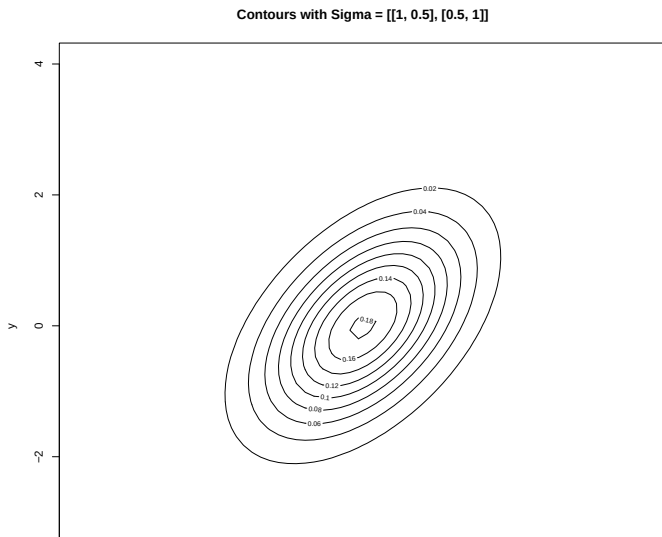
The Bivariate Normal Distribution

But what if X and Y are not independent?



The Bivariate Normal Distribution

But what if X and Y are not independent?



The Bivariate Normal Distribution

Two random variables X and Y are said to be **bivariate normal**, or **jointly normal**, if $aX + bY$ has a Normal distribution for all $a, b \in \mathbb{R}$.

The Standard Bivariate Normal Distribution

Two random variables X and Y are said to have the **standard bivariate normal distribution with correlation coefficient ρ** if their joint PDF is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left\{-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)\right\},$$

where $\rho \in (-1, 1)$. If $\rho = 0$, then we just say X and Y have the **standard bivariate normal distribution**.

The Standard Bivariate Normal Distribution

Theorem: Let X and Y be two bivariate normal random variables. Then there exist independent standard normal random variables Z_1 and Z_2 such that

$$X = \sigma_X Z_1 + \mu_X$$

$$Y = \sigma_Y \left(\rho Z_1 + \sqrt{1 - \rho^2} Z_2 \right) + \mu_Y.$$

The Standard Bivariate Normal Distribution

Two random variables X and Y are said to have the **bivariate normal distribution with parameters $\mu_X, \sigma_X^2, \mu_Y, \sigma_Y^2$ and ρ** if their joint PDF is given by

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \times \exp \left\{ -\frac{1}{2} \left[\left(\frac{x-\mu_X}{\sigma_X} \right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 - 2\rho \left(\frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} \right) \right] \right\}$$

where $\mu_X, \mu_Y \in \mathbb{R}$, $\sigma_X, \sigma_Y > 0$ and $\rho \in (-1, 1)$ are all constants.

Conditional Expectation and Variance

Theorem: Suppose X and Y are jointly normal random variables with parameters μ_X , σ_X^2 , μ_Y , σ_Y^2 and ρ . Then, given $X = x$, Y is normally distributed with

$$E[Y \mid X = x] = \mu_Y + \rho\sigma_Y \left(\frac{x - \mu_X}{\sigma_X} \right).$$
$$V[Y \mid X = x] = (1 - \rho^2)\sigma_Y^2.$$

Theorem: If X and Y are bivariate normal and uncorrelated, then they are independent.

The Multivariate Normal Distribution

But we can take this further!

The random variables $X_1, X_2, \dots, X_n$ have a **multivariate normal distribution** if their joint PDF is given by:

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} (\tilde{x} - \tilde{\mu})^T \Sigma^{-1} (\tilde{x} - \tilde{\mu}) \right\},$$

where $\tilde{\mu} = [\mu_1, \mu_2, \dots, \mu_n]^T$ is a vector of the means, and Σ is the covariance matrix.

The Multivariate Normal Distribution

The marginals of a joint normal distribution is normal.

The conditional distributions of a joint normal distribution is normal.

You can think of linear regression as a joint normal distribution from the independent errors.